

Software Requirements Specification

for

Chithara AI music generator

Version 1.2

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February 7, 2026

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Table of Contents

Table of Contents.....	ii
Revision History.....	ii
1. Introduction.....	3
1.1 Purpose	3
1.2 Document Conventions.....	3
1.3 Intended Audience and Reading Suggestions.....	4
1.4 Product Scope	5
1.5 References.....	6
2. Overall Description.....	7
2.1 Product Perspective	7
2.2 Product Functions.....	7
2.3 User Classes and Characteristics.....	8
2.4 Operating Environment.....	9
2.5 Design and Implementation Constraints.....	9
2.6 Assumptions and Dependencies	10
3. Stakeholders	11
3.1 Stakeholder Identifications and Roles	11
4. User Stories	11
4.1 Registered User Stories	11
4.2 Guest User Stories	12
5. External Interface Requirements	12
5.1 User Interface	12
5.2 Software Interface	13
5.3 Communications Interfaces	13
6. System Features	14
6.1 User Authentication	14
6.2 Generation & Quota	14
6.3 Library Management	14
6.4 Playback & Export	15
6.5 Sharing & Privacy	15
7. Other Nonfunctional Requirements	15
7.1 Safety Requirements	15
7.2 Security Requirements	15
7.3 Software Quality Attributes	16
7.4 Business Rules	16
8. Use Case Model	16
Appendix A: Glossary	19
Appendix B: To Be Determined List	19

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document identifies the functional and non-functional requirements for the Chithara AI music generator, Version 1.0. The document details the functional and non-functional requirements for the platform, which aims to revolutionize personal music creation by leveraging Artificial Intelligence to transform user prompts into high-quality audio tracks, providing structured library management, and facilitating secure content sharing.

This SRS will serve as the foundation for the subsequent system design and development phases, ensuring that all stakeholders—including developers, UI/UX designers, and project managers—have a clear understanding of the system’s capabilities and operational workflows.

1.2 Document Conventions

This document follows the conventions defined in the tables below to ensure clarity and consistency across the requirements for the Chithara AI music generator.

Term	Description
SHALL	A mandatory requirement. The feature must be implemented in Version 1.0 for the system to be considered complete.
SHOULD	A high-priority requirement that is planned for Version 2.0. Architecture should support this for future scalability.
MAY	An optional or "nice-to-have" feature. Implementation is at the discretion of the development team as time permits.
TBD	To Be Determined; information that is currently unknown (e.g., specific API costs or model names) and will be updated later
Note	Provides additional context, clarification, or examples for a specific requirement.

Requirements are categorized as follows:

Requirement Number	Description
FR-XX	Functional Requirements: Features like AI generation, library management, and playback.
NFR-XX	Non-Functional Requirements: Quality attributes such as performance, reliability, and usability
US-XX	User Stories: High-level statements describing user goals and needs from the perspective of different stakeholder roles.
UC-XX	Use Cases: Step-by-step scenarios describing user-system interactions (e.g., "UC-02: Generate Music").

1.3 Intended Audience and Reading Suggestions

This SRS is intended for all stakeholders involved in the design, development, and implementation of the **Chithara AI music generator**. Primary stakeholders include the client (Khunakorn PATTAYAKORN), the project instructor (Sirisilp KONGSILP), and the technical team (developers, testers, and maintainers).

Document Organization:

The document is organized as follows:

- **Section 1 and 2:** Provide a high-level overview of the Chithara platform, its business goals, user classes, and general constraints.
- **Section 3:** Defines the external interfaces, including Google OAuth for login and the external AI Service API for music generation.
- **Section 4:** Contains the detailed Functional Requirements (FR) organized by system feature.
- **Section 5:** Details the Non-functional Requirements, such as performance, security protocols, and business rules regarding quotas.
- **Appendices:** Include a Glossary (A), Analysis Models/Use Cases (B), and a list of pending items (C).

Suggested Reading Sequence:

To ensure the most efficient use of this document, readers should begin with **Section 2 (Overall Description)** to understand the project context and then follow the paths below:

- **Project Managers and Product Owners:** Should focus on **Section 1**, **Section 2**, and **Section 5.5 (Business Rules)** to ensure the product scope and weekly generation limits align with business objectives.
- **Developers:** Should prioritize **Section 3 (External Interfaces)** and **Section 4 (System Features)**. They should also refer to **Appendix B (Use Case Model)** to understand the logic flow for AI generation and sharing.
- **Testers (QA):** Should focus on **Section 4 (System Features)** and the step-by-step success scenarios in **Appendix B (Use Cases)** to

develop comprehensive test plans and validation scripts.

1.4 Product Scope

The Chithara AI music generator is a web-based platform designed to democratize music production by allowing users to generate high-quality, personalized audio tracks through the power of Artificial Intelligence. By providing a simple interface where users can input text-based descriptions—such as title, occasion, genre, and mood—Chithara eliminates the technical barriers to music creation for non-musicians while providing a robust management system for enthusiasts.

Major Objectives and Goals:

- **Automated Audio Generation:** To provide an interface for mapping user inputs (Title, Occasion, Genre, Mood) into prompts for an AI backend to generate complete audio tracks.
- **Hierarchical Asset Management:** To implement a cloud-based relational database where users can perform CRUD (Create, Read, Update, Delete) operations on songs and group them into logical "Albums."
- **Multi-Tiered Access Control:** To develop a security framework that manages data visibility through three distinct states: Public (Global), Private (User-only), and Exclusive (Authenticated via unique email invitations).
- **System Resource Throttling:** To maintain server stability and cost-efficiency by implementing a metered "Generation Quota" system that resets on a weekly cycle.

In-Scope Features (Functional Boundaries):

- **User Account Management:** Secure registration, login, and profile management.
- **AI Request Interface:** A front-end form to capture metadata and a backend handler to communicate with the generative AI model.
- **Cloud Library:** A dashboard for playback, metadata editing, and organizing tracks into albums.
- **Sharing Engine:** Generation of unique URLs for public sharing and an automated email dispatch system for "Exclusive" access invitations.
- **Quota Tracker:** A backend logic module to track user generation counts and restrict access once the weekly limit is reached.

Out-of-Scope Features (System Limitations):

- **Manual Audio Editing:** The system will not include tools for trimming, fading, mixing, or multi-track manipulation of the generated files.
- **User Audio Uploads:** Users cannot upload their own MP3s or vocals to be "remixed" by the AI; the system only supports text-to-audio generation.
- **Monetization/Marketplace:** The system will not handle financial transactions, digital rights management (DRM) licensing, or a marketplace for selling tracks.
- **Real-time Collaboration:** There is no support for "Live" shared editing sessions where two users work on the same album.

simultaneously.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society.
- Google Identity Platform, Google OAuth 2.0 Documentation. Available at: <https://developers.google.com/identity/protocols/oauth2>
- Course Project Assignment, "Cithai Music Generator" Scenario (provided by Sirisilp KONGSILP).
- External AI Service API Documentation, Use Suno API for music synthesis integration.

2. Overall Description

2.1 Product Perspective

The Chithara AI music generator is a standalone web-based system designed to **facilitate the generation and management of AI-synthesized audio files**. While the platform operates initially as a music generation tool, it is architected to evolve into a comprehensive creative audio hub featuring expanded social sharing and advanced digital asset management capabilities.

The system is designed for cross-platform accessibility, functioning seamlessly on both desktop and mobile web browsers. It acts as an orchestrator between the user and several high-level external services to deliver a streamlined creative experience.

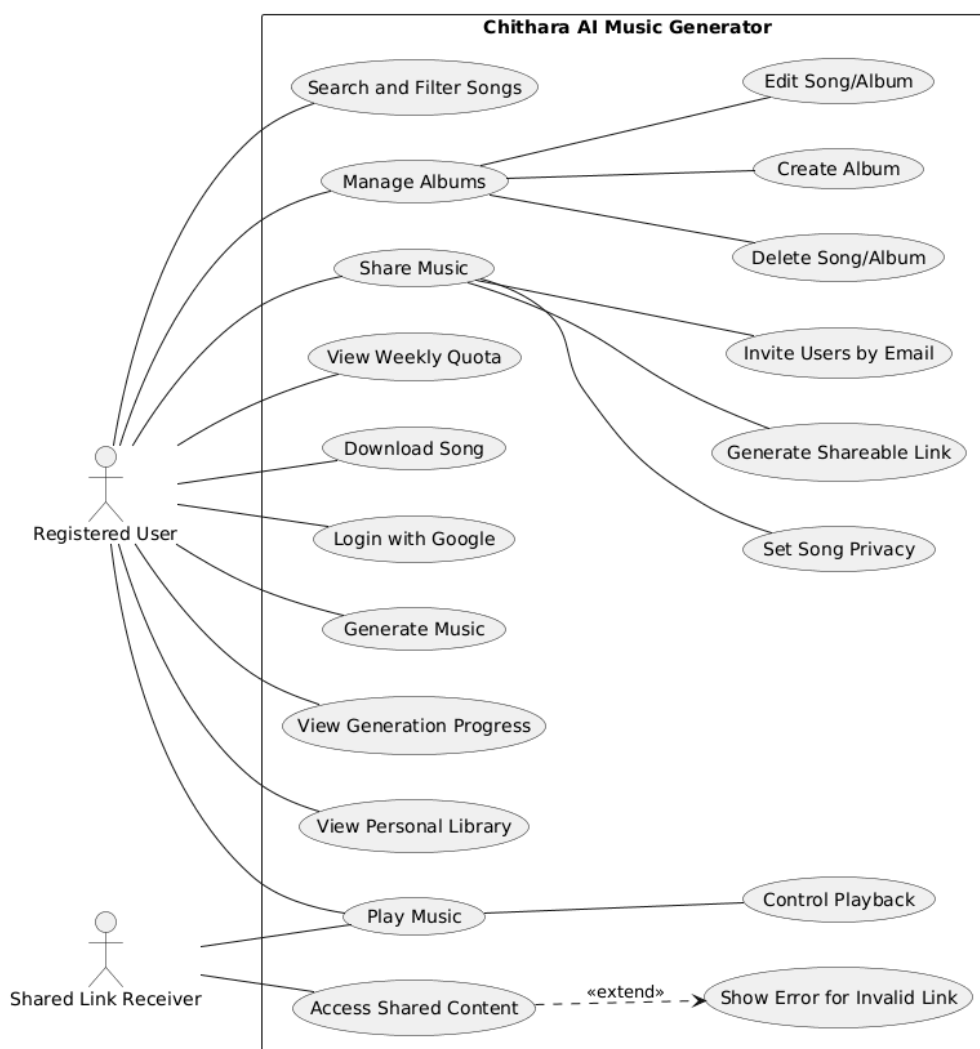
2.2 Product Functions

The Chithara AI music generator provides a comprehensive set of functions to enable users to create, organize, and share AI-generated music. These functions are summarized into the following major groups:

1. **Authentication:** Secure Google OAuth 2.0 login and profile management .
2. **AI Music Synthesis:** A prompt-based interface for title, genre, and mood, including a confirmation step and real-time progress indicators.
3. **Library Management:** Automatic cloud saving, album/folder organization, and local MP3 downloads .
4. **Media Player & Navigation:** Full-featured audio playback controls and library search/filtering by genre or occasion .
5. **Sharing & Privacy:** Toggle-based privacy settings (Public/Private), shareable link, and secure email-based invitations with guest access control.

6. **Quota Enforcement:** A monitoring system to track and display remaining weekly generation limits, preventing over-usage of AI resources

Use Case Diagram of the Chithara AI Music Generation System :



2.3 User Classes and Characteristics

The platform will serve the following user classes:

- **End Users (Registered Users):** Hobbyists and creators who use the AI to generate music. They require an intuitive interface that works on both mobile and desktop.
- **Unauthenticated Users (Shared Link Receivers):** External contacts who play shared content. Their experience must be friction-less, requiring no account to play "Public" content.
- **System Administrators / Maintainers:** The technical team responsible for API health, cloud storage management, and security. They require high-level access to logs and system configurations.

2.4 Operating Environment

The platform will operate in the following environment:

- **Client-Side Environment:**
 - Browsers: Optimized for modern versions of Chrome, Safari, Firefox, and Edge.
 - Devices: Fully responsive design for Desktop devices.
- **Server-Side Environment:**
 - Cloud Infrastructure: Hosted on cloud platforms (e.g., AWS, GCP, or Vercel) to support audio streaming and API orchestration.
 - Database: A scalable system (PostgreSQL/MongoDB) to manage metadata and user relationships.
 - Storage: Secure cloud buckets (e.g., AWS S3) for MP3 file hosting.
- **External Integrations:**
 - Authentication: Google Identity Services API.
 - Music Generation: 3rd-party Neural Network Music Synthesis API.
- **User Connectivity:** The system is designed to handle varying internet speeds, with the audio player optimized for streaming on moderate bandwidth.

2.5 Design and Implementation Constraints

The following constraints will impact the design and implementation of the Chithara AI music generator platform:

2.5.1 Technical Constraints

- **Web-Based Architecture:** The system must be developed as a responsive web application accessible via URL on both desktop and mobile browsers, requiring no local installation.
- **External API Integration:** Core functionality depends on a 3rd-party AI Music Synthesis API. The system must be architected to handle external service latency, rate limits, and potential changes in API data structures.
- **Authentication Standard:** The application will use Google OAuth 2.0 as the primary authentication mechanism to ensure security and reduce password management overhead. The system will not store user passwords internally. The authentication design allows for future

extension to alternative authentication providers if required.

2.5.2 Resource and Sustainability Constraints

- **Generation Quotas:** To manage API costs for the client, the system must enforce a configurable weekly limit on the number of tracks a user can generate.
- **Storage Management:** The implementation must include constraints on file size and audio bitrate to ensure cloud storage usage remains within the project's budgetary limits.

2.5.3 Security and Privacy Constraints

- **Access Control:** The system must ensure that tracks marked as "Private" are inaccessible via direct URL to any unauthenticated user or anyone not explicitly invited by the owner.

2.5.4 Language and Tools

- **Language:** The user interface, system notifications, and all technical documentation must be provided in English.
- **Frameworks:** Developers must utilize modern, well-supported frameworks (e.g., React, Node.js, or Python) to ensure the system is scalable and easily manageable for future maintainers.

2.6 Assumptions and Dependencies

2.6.1 Assumptions

- **External Service Uptime:** *It is assumed that the 3rd-party AI Music Synthesis API is functional and maintains high availability. The system's core value proposition is dependent on this external service's uptime.*
- **Continuous Session Persistence:** *It is assumed that the user will maintain an active browser session (keep the tab open) during the initial handshake and initialization phase of the music generation process to ensure the request is successfully registered and tracked by the frontend.*

2.6.2 Dependencies

- **Network Connectivity:** *The system is entirely dependent on the user having a stable and active internet connection. Both the generation of new tracks and the streaming of existing audio from the cloud library require consistent bandwidth.*
- **Google Identity Provider:** *The ability to authenticate users is dependent on the availability of Google's OAuth 2.0 servers.*

3. Stakeholders

3.1 Stakeholder Identification and Roles

Stakeholder	Role / Responsibility
Client / Product Owner	Defines system vision, approves scope, and validates requirements.
End Users (Registered Users)	Primary end users who generate, manage, and share AI-generated music
Unauthenticated Users (Shared Link Receivers)	Secondary users who access shared music via provided links or invitation link but cannot generate or manage content
System Administrators (Non End User)	Responsible for deployment, configuration, monitoring, and managing API usage limits. This role does not interact with the application's user-facing features.

4. User Stories

This section describes the primary user needs of the Chithara AI Music Generator system. User stories are organized by stakeholder role to ensure clarity, coverage, and traceability to system features and functional requirements.

4.1 Registered User Stories

- **US-1:** As a registered user, I want to log in using my Google account so that my generated music is securely saved and accessible across devices.
- **US-2:** As a registered user, I want to describe a song using a title, genre, mood, and occasion so that the AI can generate music that matches my intent.
- **US-3:** As a registered user, I want to review song details before generation so that I can avoid wasting my generation quota.
- **US-4:** As a registered user, I want to see visual feedback during music generation so that I know the system is processing my request.
- **US-5:** As a registered user, I want generated songs to be automatically saved to my personal library so that I can listen to them later.
- **US-6:** As a registered user, I want to create and manage albums so that I can organize related songs.
- **US-7:** As a registered user, I want to control music playback (play, pause, skip, rewind and fast-forward) so that I can easily listen to my content.
- **US-8:** As a registered user, I want to search and filter my library so that I can quickly find specific songs.
- **US-9:** As a registered user, I want to set privacy levels (public or private) for my songs so that I control who can access them.
- **US-10:** As a registered user, I want to share songs or albums via links or email invitations so that selected people can listen to my music.
- **US-11:** As a registered user, I want to see my remaining weekly generation quota so that I can manage my usage.
- **US-12:** As a registered user, I want to download my generated songs so that I can listen to them offline.

- **US-13:** As a registered user, I want to see a clear error message when a shared link is invalid or unauthorized so that I understand why access failed.
- **US-14:** As a registered user, I want to view all songs within an album so that I can easily select and play a specific song.
- **US-15:** As a registered user, I want to edit or delete my songs and albums so that I can correct mistakes or remove unwanted content.

4.2 Guest User Stories

- **US-16:** As a guest user, I want to access shared music using a link so that I can listen without creating content.
- **US-17:** As a guest user, I want to log in using Google when required so that my identity can be verified for restricted content.

5. External Interface Requirements

5.1 User Interfaces

The Chithara user interface (UI) will be a responsive, web-based dashboard designed for simplicity and ease of use across desktop devices.

5.1.1 General Design Principles

- **Responsive Layout:** The UI must automatically adjust for screen size, prioritizing a "Desktop-First" approach for complex music generation prompts.
- **Loading States:** Since AI generation takes time, the UI must provide clear visual feedback (progress bars, spinners, or status messages) to inform the user that a track is being processed.

5.1.2 Layout Structure

The interface will follow a standard dashboard layout consisting of three main zones:

1. **Navigation Bar/Sidebar:** Contains links to the Music Library, AI Generator, and Profile Settings. It must be accessible from every screen.
2. **Main Workspace:** The central area where the user interacts with the music generation form, browses their library, or manages albums.
3. **Persistent Music Player:** A global audio control bar located at the bottom of the screen. It remains visible while the user navigates different pages, allowing for uninterrupted playback.

5.1.3 Key UI Components

- **Authentication Screen:** Features a prominent "Sign in with Google" button as the primary entry point.
- **Generation Interface:** A clean form containing input fields for track title, genre, mood, and occasion, followed by a "Generate" confirmation button.

- **Music Library:** A grid or list view displaying track metadata (title, date, duration) with quick-action buttons for "Share," "Download," and "Edit."
- **Sharing Modal:** A pop-up interface that allows users to toggle "Public/Private" settings and copy a shareable URL or enter an email address for invitations.

5.1.4 Error and Feedback Standards

- **Error Messages:** System errors (e.g., API timeouts or invalid links) must be displayed as non-intrusive "Toast" notifications or red alert text near the relevant action button.
- **Quota Alerts:** A visual indicator must show the user's remaining weekly generation credits, with a clear warning when the limit is reached.

5.2 Software Interfaces

- **AI Backend API:** The system shall communicate via **REST/JSON** to send user prompts (Title, Genre, Mood, Occasion). It shall receive either a direct audio stream or a temporary URL to the generated MP3 asset.
- **Database Interface:** The system shall perform CRUD operations for song metadata.

5.3 Communications Interfaces

- All data transfer shall occur over HTTPS.
- The system shall support real-time to update song status (e.g., from "Generating" to "Ready").

6. System Features

6.1 User Authentication

6.1.1 Description

Manages secure entry into the platform for both creators and guest listeners.

6.1.2 Functional Requirements

- **FR-1:** The system shall allow users to log in using their Google account.
- **FR-2:** The system shall store user identity information retrieved from Google authentication.
- **FR-3:** The system shall allow shared-link receivers to log in using Google to access shared content.

6.2 AI Music Generation & Quota Management

6.2.1 Description

The core engine allows users to synthesize music based on metadata while managing resource limits.

6.2.2 Functional Requirements

- **FR-4:** The system shall allow users to input song details including title, occasion, genre, and mood.
- **FR-5:** The system shall allow users to review and confirm song details before generation.
- **FR-6:** The system shall generate music based on user-provided descriptions using AI.
- **FR-7:** The system shall display a loading indicator while generation is in progress.
- **FR-8:** The system shall display the number of remaining music generations available for the current week.

6.3 Library Management

6.3.1 Description

Provides tools for permanent storage, organization, and local portability of music.

6.3.2 Functional Requirements

- **FR-9:** The system shall automatically save generated songs to the user's library.
- **FR-10:** The system shall allow users to create albums and view songs within them.
- **FR-11:** The system shall allow users to edit or delete songs and albums from their library .
- **FR-12:** The system shall allow users to download generated songs to their local device.

6.4 Playback, Search & Navigation

6.4.1 Description and Priority

The interface for interacting with and discovering content within the library.

6.4.2 Functional Requirements

- **FR-13:** The system shall provide playback controls: play, pause, skip, rewind, and fast-forward.
- **FR-14:** The system shall allow users to search songs by title.
- **FR-15:** The system shall allow users to filter songs by genre or occasion.

6.5 Sharing & Privacy Control

6.5.1 Description and Priority

Manages visibility permissions and secure distribution of music.

6.5.2 Functional Requirements

- **FR-16:** The system shall allow users to set song privacy to Public or Private.
- **FR-17:** The system shall generate shareable links for public content.
- **FR-18:** The system shall allow sharing via email invitations.
- **FR-19:** The system shall restrict access to private songs to their owners only.

- **FR-20:** The system shall display an error message for invalid or unauthorized shared links.

7. Other Nonfunctional Requirements

7.1 Safety Requirements

- **NFR-1: Content Moderation:** The system must actively prevent the generation of hate speech, discriminatory content, or illegal themes via automated input validation of the user's prompts.
- **NFR-2: Data Integrity:** In the event of a generation failure, the system must ensure the user's weekly quota is not deducted

7.2 Security Requirements

- **NFR-3: Token Security:** As the system uses Google OAuth 2.0, all authentication tokens must be stored securely (e.g., HttpOnly cookies) and never exposed in plain text.
- **NFR-4: Credential Protection:** API Keys for the 3rd-party AI Service and Google Client Secrets must be stored in server-side environment variables and never exposed to the client-side browser.

7.3 Software Quality Attributes

- **Reliability:** The system shall implement an automatic retry mechanism for transient network errors when communicating with the AI Synthesis API.
- **Compatibility:** The application must be fully responsive and function correctly on modern versions of Chrome, Safari, Firefox, and Edge (Desktop and Mobile).
- **Usability:** The interface must adhere to a "Three-Click Rule," ensuring users can reach the generation screen or play a song from the library in three clicks or fewer.

7.4 Business Rules

- **NFR-1: Pricing Model:** The application is currently free to use; no payment gateway integration is required for this phase.
- **NFR-2: Resource Sustainability (Quota):** Each registered user is restricted to a set number of music generations per week (as defined by the Client) to manage API operational costs.
- **NFR-3: Privacy by Default:** All newly generated songs are set to "Private" by default. Visibility is only changed to "Public" or "Shared" through explicit user action.
- **NFR-4: Identity Requirement:** Access to music generation and personal library features is strictly reserved for users with a verified Google account.

8. Use Case Model

This section describes the use case model of the Chithara AI Music Generator. The use case diagram is presented in **Section 2.2 (Product Functions)**. This section focuses on use case descriptions.

UC-01: Log in with Google

Primary Actor: Registered User

Precondition: User is not currently logged in.

Postcondition: User is authenticated and redirected to their personal dashboard/library.

Main Success Scenario:

1. The user opens the AI Music Generation System.
2. The system displays the login page with a “Login with Google” option.
3. The user clicks the “Login with Google” button.
4. The system redirects the user to the Google authentication service.
5. The user enters their Google account credentials.
6. Google authenticates the user successfully.
7. Google returns authentication confirmation and user identity data to the system.
8. The system validates the authentication response.
9. The system creates or retrieves the user’s account profile.
10. The system establishes a user session.
11. The system redirects the user to their personal dashboard or music library.
12. The system displays a confirmation that login was successful.

UC-02: Generate Music

Primary Actor: Registered User

Precondition: User is logged in and has remaining generation quota.

Postcondition: A new AI-generated song is created and saved to the user’s library.

Main Success Scenario:

1. The user navigates to the music generation page.
2. The system displays a form for entering song details.
3. The user enters the song title, occasion, genre, and mood description.
4. The user submits the song details.

5. The system displays a review screen with the entered details.
6. The user confirms the song details.
7. The system starts the AI music generation process.
8. The system displays a loading indicator to show generation progress.
9. The AI service successfully generates the music.
10. The system receives the generated music file.
11. The system saves the generated song to the user's personal library.
12. The system updates the user's remaining generation quota.
13. The system displays a success message and provides access to play the new song.

UC-03: Manage Albums

Primary Actor: Registered User

Precondition: User is logged in.

Postcondition: The user's albums are updated according to the performed action.

Main Success Scenario:

1. The user navigates to the album management section.
2. The system displays the user's existing albums.
3. The user selects the option to create or manage an album.
4. The system displays album management options.
5. The user creates a new album or selects an existing album.
6. The user adds one or more songs to the album.
7. The system saves the album changes.
8. The system updates the album contents in the user's library.
9. The system displays the updated album to confirm the changes.

UC-04: Access Shared Content

Primary Actor: Shared Link Receiver

Precondition: The user has a valid shared link.

Postcondition: The user gains access to the shared song or album.

Main Success Scenario:

1. The user clicks a shared song or album link.
2. The system opens the shared content access page.
3. The system prompts the user to log in using Google if not already authenticated.
4. The user logs in using their Google account.
5. The system validates the shared link.
6. The system verifies that the user is authorized to access the shared content.
7. The system retrieves the shared song or album.
8. The system displays the shared content to the user.
9. The system allows the user to play the shared music.

Appendix A: Glossary

- **AI (Artificial Intelligence):** In this context, refers to the 3rd-party neural network service used to synthesize original music tracks based on user prompts.
- **API (Application Programming Interface):** A set of protocols used for Chithara to communicate with external services (Google and the AI Music Generator).
- **Google OAuth 2.0:** The industry-standard authorization protocol used to allow users to securely sign in using their Google accounts.
- **HTTPS (Hypertext Transfer Protocol Secure):** The encrypted communication protocol used to protect data in transit between the user and the server.
- **Prompt:** The descriptive user input (genre, mood, occasion) used to guide the AI in generating a specific music track.
- **Quota:** The weekly limit enforced by the system to restrict the number of songs a user can generate, ensuring budget sustainability for the client.
- **SRS (Software Requirements Specification):** This document, which outlines the features and constraints of the Chithara application.
- **WCAG (Web Content Accessibility Guidelines):** International standards for web accessibility, ensuring the platform is usable by people with disabilities.

Appendix B: To Be Determined List

- *Selection of AI Synthesis Vendor (TBD-1): The specific 3rd-party provider for the music generation neural network (e.g., Suno, Udio, or a generic API) is to be finalized based on latency and cost-per-generation.*
- *Weekly Generation Quota (TBD-2): The exact number of music tracks a user is permitted to generate each week must be defined by the Client to align with the operational budget.*